# M13D — Mock exam 1

Instruments · Rotary Wing Aerodynamics · Autoflight · Equipment and Furnishings

40 questions · 20 Instruments · 13 Autoflight · 5 Rotary Wing Aerodynamics · 2 Equipment and Furnishings

Level 3: 22 · Level 2: 13 · Level 1: 5. Three options, one correct.

Every question sits on a fact highlighted or underlined in the manual. No question in this paper appears in any of the other four.

The answer key follows the questions, with the explanation and the manual PDF page for each one.

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## INSTRUMENTS

**1.** A crew flies at a level that lies inside the troposphere over the equator but above it over the poles. Which thicknesses of the troposphere account for this?

- **A.** About 8.5 km at the equator and about 16.8 km at the poles
- **B.** About 16.8 km at the equator and about 8.5 km at the poles
- **C.** About 16.8 km at the equator and about 11 km at the poles

**2.** A fleet of six aircraft is to be cleared for RVSM. Who must apply for the Airworthiness Approval and for what?

- **A.** The aircraft operator, for the whole fleet at once
- **B.** The aircraft manufacturer, for each individual aircraft
- **C.** The aircraft operator, for each individual aircraft

**3.** An aeroplane extends its flaps on the approach. Which reading can change as a result of this alone?

- **A.** The total air temperature reading
- **B.** The static pressure reading
- **C.** The angle of attack reading

**4.** An aircraft climbs. What happens inside its barometric altimeter?

- **A.** Pressure decreases, the bellows contract and the pointer rises
- **B.** Pressure decreases, the bellows expand and the pointer rises
- **C.** Pressure increases, the bellows expand and the pointer falls

**5.** One of the factors on which the choice of altimeter depends is

- **A.** The number of static ports fitted
- **B.** The tolerance values required by law
- **C.** The transition level at the destination

**6.** An altimeter is left calibrated to 1013 hPa while the real pressure at sea level below the aircraft is lower. What will the instrument indicate?

- **A.** Too high an altitude
- **B.** The correct altitude
- **C.** Too low an altitude

**7.** A jet aeroplane climbs from low altitude to high altitude. Which limit governs its maximum speed at the higher altitude?

- **A.** The never exceed speed VNE
- **B.** The maximum allowed Mach number MMO
- **C.** The maximum operating velocity VMO

8. A fast jet aeroplane cannot use the simple OAT indicating system. What is the reason?
   - A. It needs a bleed air supply that jets cannot provide
   - B. It cannot be protected against ice
   - C. It is only accurate up to an airspeed of 250 kts

9. The temperature supply to an analogue air data computer is lost. Which outputs can no longer be produced?
   - A. Calibrated airspeed and altitude
   - B. Vertical speed and total air temperature
   - C. True airspeed and static air temperature

10. A GNADIRU is fitted but the aircraft has no air data computer as a separate box. How does the unit obtain its pressures?
   - A. From air data modules placed as near as possible to the pitot and static sensors
   - B. From pneumatic tubes run from the probes to the unit in the avionics bay
   - C. From the inertial reference section, which derives the air data from accelerations

11. After a deviation warning the aircraft goes on drifting away from the original altitude. At what point does the alert end?
   - A. When the aircraft is more than 900 ft from the original altitude
   - B. When the aircraft is more than 300 ft from the original altitude
   - C. When the aircraft reaches the altitude selected on the control panel

12. With which unit does the solid state flight recorder have a direct interface
   - A. the centralised data acquisition module
   - B. the flight warning computer
   - C. the flight management computer

13. When is the recorded data of the wireless quick access recorder compressed, encrypted and transmitted
   - A. while the aircraft is in the cruise
   - B. while the aircraft is being powered down
   - C. while the aircraft is on the ground

14. An exterior force has been applied to the axis of a running gyro and is held there. When does the gyro stop precessing?
   - A. when its axis has turned through a right angle from the direction it started in
   - B. when its sense of rotation corresponds to the rotational direction acting upon it
   - C. when the inertia of its rotating mass has fully absorbed the applied force

15. A unit on the bench is labelled as a directional gyro. What does that tell a technician about the way its spin axis lies?
   - A. it is vertical
   - B. it is horizontal
   - C. it is at 45 degrees

16. In a direct reading compass the damping medium changes volume with temperature. What compensates for this?
   - A. Bellows
   - B. Slots
   - C. Washers

17. A remote compass system is fitted to an aircraft. Which two sensing devices does it use?
    A. a sensor for the earth's magnetic field and a vertical gyro
    B. a sensor for the aircraft's own field and a rate gyro
    C. a sensor for the earth's magnetic field and a directional gyro

18. How does the master warning light behave for a level 3 warning
    A. It flashes
    B. It stays steady
    C. It stays dark

19. An aircraft penetrates the mode 4A boundary at 250 knots with the landing gear still retracted. Which message is generated?
    A. TOO LOW TERRAIN
    B. TOO LOW GEAR
    C. TOO LOW FLAPS

20. An operator does not want the TWENTY FIVE HUNDRED callout at 2500 feet. Which callout can be selected instead?
    A. The MINIMUMS MINIMUMS callout
    B. The RADIO ALTITUDE callout
    C. The TOO LOW TERRAIN callout

## AUTOFLIGHT

21. The input of a D-regulator changes at a constant rate. What does the output value do?
    A. It increases linearly without limit
    B. It falls back towards zero
    C. It remains steady at one level

22. A quantity describing a linear motion of the aircraft is to be regulated. Which part of the autopilot has this task?
    A. The outer loops, since the quantity is a flight condition parameter
    B. The inner loop, since the quantity is a flight path parameter
    C. The outer loops, since the quantity is a flight path parameter

23. The roll channel of the autopilot/flight director is working as an independent subsystem. Why can it not guide the aircraft along a horizontal profile?
    A. It needs the pitch channel to be engaged in a vertical profile mode
    B. It needs inputs from the pilot or from a flight management computer system
    C. It needs the auto-thrust system to set a flight phase thrust rating

24. The crew must leave the FMS profile at short notice after an air traffic control request, but wishes to keep the autopilot flying. Which level of guidance do they use?
    A. Selected manual
    B. Selected automatic
    C. Managed automatic

25. The pilot pushes firmly on the control column while the autopilot is engaged. What happens to the auto-trim function?
    A. It is transferred to the elevator actuator
    B. It is deactivated until the force is removed
    C. It is switched over to its high trim speed

**26.** The flight management system requests a climb. Why must the autoflight system have a pilot input before it starts that climb?

- **A.** To prevent the autothrust system reverting to thrust mode
- **B.** To prevent the flight management system losing its reference
- **C.** To prevent altitude changes being made without ATC clearance

**27.** An aircraft tracking a radial passes overhead the VOR station. Why does the VOR mode need a separate over station phase?

- **A.** Because the deviation signal is useless in the cone of confusion
- **B.** Because the receiver frequency must be re-tuned as the station is passed
- **C.** Because the pre-selected course must be reversed as the station is passed

**28.** Why is the yaw alignment left until shortly before touchdown

- **A.** It minimizes the lateral drift
- **B.** It minimizes the crosswind component
- **C.** It minimizes the rudder deflection

**29.** The reported runway visual range is 100 m and the decision height is 40 ft — which category covers it?

- **A.** CAT III a
- **B.** CAT III b
- **C.** CAT III c

**30.** An approach is flown to a decision height of 250 ft with a reported visual range of 600 m — which category applies?

- **A.** CAT I
- **B.** CAT II
- **C.** CAT III a

**31.** An airport has no runway visual range measuring equipment — which value may be converted instead?

- **A.** The meteorological visibility
- **B.** The cloud base height measured
- **C.** The decision height selected

**32.** A low visibility take-off is a take-off made below a runway visual range of

- **A.** 400 metres
- **B.** 550 metres
- **C.** 300 metres

**33.** One of several parallel channels drops out and the output signal is unaffected — which redundancy is in use?

- **A.** Monitored redundancy
- **B.** Passive redundancy
- **C.** Active redundancy

## ROTARY WING AERODYNAMICS

**34.** What is the leading edge of a rotor blade

- **A.** The edge opposite the one that cuts into the wind
- **B.** The outer end of the blade opposite the root
- **C.** The edge that cuts into the wind as the blade turns

**35.** Pitch angle and angle of attack are measured from different references. Which pairing of references is correct?

- **A.** Pitch angle is referenced to the axis of rotation, angle of attack to the span line
- **B.** Pitch angle is referenced to the plane of rotation, angle of attack to the relative wind
- **C.** Pitch angle is referenced to the relative wind, angle of attack to the plane of rotation

**36.** On a turning blade the relative wind does not lie in the direction of the plane of rotation. What follows for the pitch angle and the angle of attack?

- **A.** They cannot be equal to each other
- **B.** They must change at the same rate
- **C.** They must be equal to each other

**37.** A pilot pushes the cyclic stick forward to fly forward. Which chain of events takes place at the swash plate and the blades?

- **A.** The swash plate is lifted, the control rods move and the blade pitch rises on every blade at once
- **B.** The swash plate turns, the bell cranks move and the blade pitch is held through the revolution
- **C.** The swash plate tilts, the control rods move and the blade pitch varies through the revolution

**38.** A helicopter flies forward at a steady speed. Why can its blades not all produce the same lift?

- **A.** The airstream raises the pitch angle on one half of the disc and lowers it on the other
- **B.** The airstream increases the rotational speed of the whole rotor disc
- **C.** The airstream adds to the speed of one half of the disc and subtracts from the other

## EQUIPMENT AND FURNISHINGS

**39.** A reading light and a night illumination light both fail in the same cabin section. Why are two different subsystems involved?

- **A.** Reading lights belong to the services system and night illumination to the address system
- **B.** Reading lights belong to the lighting system and night illumination to the services system
- **C.** Reading lights belong to the services system and night illumination to the lighting system

**40.** Further functions are to be added to an existing NICE installation. Which protocol makes the system easy to expand?

- **A.** The standard TCP/IP protocol
- **B.** The ARINC 429 protocol
- **C.** The spanning tree protocol

# Answer key

Answer, explanation and the manual page each question comes from

1. **B** The troposphere reaches much higher at the equator than at the poles, so the same level can be inside it in one place and above it in the other. (L3, PDF p. 365)
2. **C** Approval is per aircraft and is applied for by the operator. (L2, PDF p. 379)
3. **B** The static source error depends on the position of the flaps, and it is the static pressure reading that this error affects. (L3, PDF p. 409)
4. **B** Falling pressure around the capsule lets the bellows expand, which drives the pointer up the scale. (L2, PDF p. 449)
5. **B** Legal tolerance values are one of the three listed selection factors. (L1, PDF p. 453)
6. **A** With the reference set higher than the real sea level pressure, the instrument over-reads. (L3, PDF p. 459)
7. **B** Above a certain altitude the Mach limit MMO takes over from the dynamic pressure limit VMO. (L3, PDF p. 465)
8. **C** Its accuracy runs out at 250 kts, which is why it is limited to slow aircraft. (L2, PDF p. 489)
9. **C** TAS and SAT are the outputs that depend on the temperature being supplied to the Mach module. (L3, PDF p. 499)
10. **A** The air data section works like a digital air data computer but is fed digitally by modules mounted close to the probes and ports. (L3, PDF p. 541)
11. **A** The deviation alert ends either on regaining the altitude or on passing 900 ft away from it. (L3, PDF p. 563)
12. **A** Its direct interface is with the CDAM. (L1, PDF p. 595)
13. **C** The transfer takes place with the aircraft on the ground. (L2, PDF p. 613)
14. **B** The gyro precesses until its sense of rotation corresponds to the rotational direction acting upon it. (L3, PDF p. 619)
15. **B** Gyros with a horizontal axis are the ones called directional gyros. (L2, PDF p. 651)
16. **A** Bellows take up the expansion and contraction of the damping fluid. (L2, PDF p. 737)
17. **C** The system pairs an earth's field sensor with a directional gyro. (L1, PDF p. 757)
18. **A** The master warning light flashes for a level 3 warning. (L2, PDF p. 879)
19. **A** With the gear still up the boundary is penetrated at 190 knots or below as TOO LOW GEAR, while the boundary that rises with airspeed to 1000 feet at 250 knots gives TOO LOW TERRAIN. (L3, PDF p. 915)
20. **B** At 2500 feet the alternative to TWENTY FIVE HUNDRED is the aural RADIO ALTITUDE. (L3, PDF p. 919)
21. **C** The differential regulator reacts to rate of change, so a steady rate gives a steady output. (L2, PDF p. 72)
22. **C** Linear motion means a flight path parameter, and flight path regulation is the outer loop task. (L3, PDF p. 76)
23. **B** On its own the roll channel has nothing to steer to, so it must be fed by the pilot or the FMCS. (L3, PDF p. 82)
24. **B** Selected automatic leaves the autopilot in control while the pilot supplies the reference values himself. (L3, PDF p. 90)
25. **B** Force on the control column deactivates auto-trim for as long as it is applied. (L2, PDF p. 96)
26. **C** The pilot input stands in for the clearance, so the aircraft never changes level on its own. (L3, PDF p. 170)
27. **A** With no usable deviation overhead, the controller cannot keep steering on it and needs another rule. (L3, PDF p. 182)
28. **A** Once the crab is taken out the aircraft starts drifting sideways, so the manoeuvre is kept short. (L3, PDF p. 218)
29. **B** A decision height below 50 ft with a visual range between 200 m and 75 m falls in CAT III b. (L3, PDF p. 272)
30. **A** A decision height of not less than 200 ft together with a visual range of not less than 550 m is CAT I. (L2, PDF p. 272)
31. **A** Met visibility can be converted where no RVR equipment exists. (L2, PDF p. 272)
32. **A** Below 400 m RVR the departure counts as a low visibility take-off. (L1, PDF p. 272)
33. **C** Only active redundancy keeps the output going with no switch over. (L3, PDF p. 283)
34. **C** The leading edge cuts into the wind. (L1, PDF p. 8)
35. **B** Both use the chord, but pitch angle uses the plane of rotation and angle of attack uses the relative wind. (L2, PDF p. 18)

**36. A** The two angles are equal only when the relative wind has the same direction as the plane of rotation, so with the relative wind elsewhere they cannot be equal. (L3, PDF p. 20)

**37. C** A cyclic input tilts the swash plate, which moves the control rods and varies blade pitch through each revolution. (L3, PDF p. 44)

**38. C** The airstream caused by the speed of flight is added to the rotational speed of the advancing blade and subtracted from that of the retreating blade, so the two sides of the disc cannot make the same lift. (L3, PDF p. 46)

**39. C** Reading lights sit under the Passenger Services System while night illumination sits under the Cabin Lighting System. (L3, PDF p. 336)

**40. A** Expandability comes from the use of the standard TCP/IP protocol. (L2, PDF p. 338)